



Class: SY B.Tech.

Course: Electromagnetic Fields

Course Code: PCC-EE204

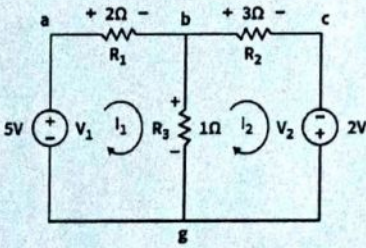
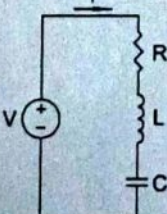
Date: 19.08.2025

Time: 11:00 A.M to 11:45 A.M

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		Question Statement	Marks	CO	BT
Q.1	a	Define the basic circuit parameters: Current and Voltage, including their units.	2	CO1	BT1
	b	Given a 10 V voltage source in series with a $5\ \Omega$ resistor, use source transformation to find the equivalent current source and the value of the parallel resistor.	3	CO1	BT3
Q.2	a	Define the following terms: (i) Node, (ii) Branch, (iii) Graph, (iv) Tree and (v) link or chords	5	CO1	BT1
Q.3	a	For the circuit shown in the figure 1, there are two voltage sources $V_1 = 5V$ and $V_2 = 2V$. Currents I_1 and I_2 pass through loops 1 and 2 respectively with resistances $R_1 = 2\ \text{ohms}$, $R_2 = 3\ \text{ohms}$ and $R_3 = 1\ \text{ohms}$. Using KVL, find currents I_1 and I_2.  Figure. 1	6	CO2	BT3
	b	The given figure 2 represents the RLC circuit. Find the output voltage of the given circuit in s-domain. (Assuming the initial conditions are 0.)  Figure 2.	4	CO2	BT3

== ALL THE BEST ==



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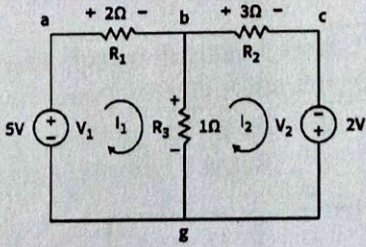
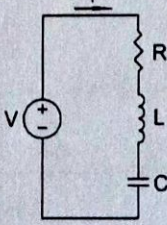
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=== ALL THE BEST ===



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: SY

Course: Network Analysis/
Circuit Theory

Course Code: PCC-
EE203/PCCIN203

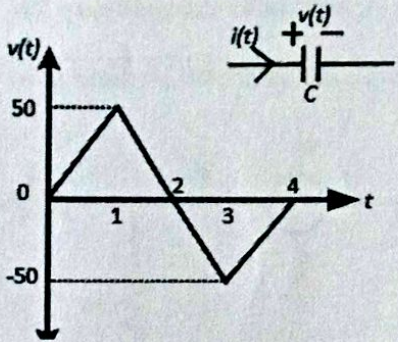
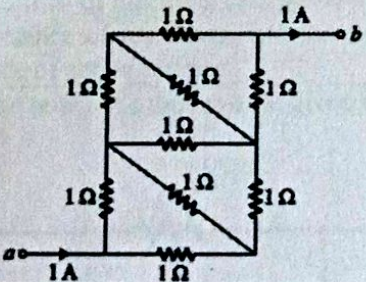
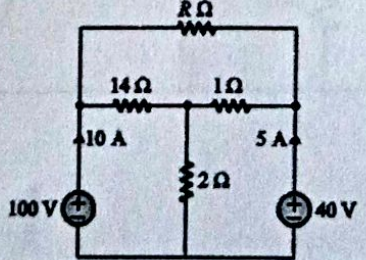
Date: 21.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

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Q.N.		Marks	CO	BT
Q.1	Sketch the waveform of $i(t)$ along with the proper notations and x and y axis. 	5	CO1	BT3
Q.2	An inductor has a linear change in current from 50mA to 100mA in 2 msec and induces a voltage of 160mV. Calculate the value of Inductor.	5	CO1	BT3
Q3	A resistor network is shown in the figure with all the values of 1Ω. What is voltage between terminal a and b. 	5	CO1	BT3
Q.4	Find the value of R in Ohms 	5	CO1	BT3

== ALL THE BEST ==

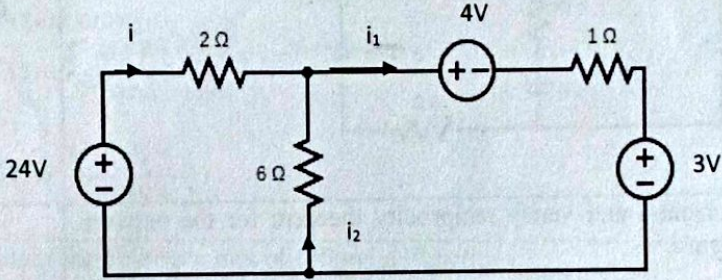
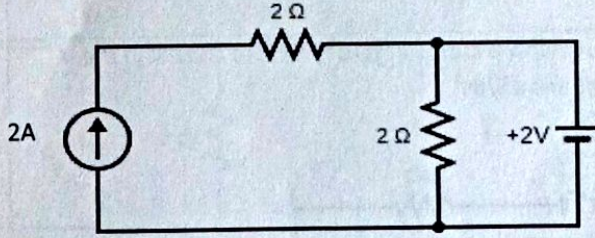


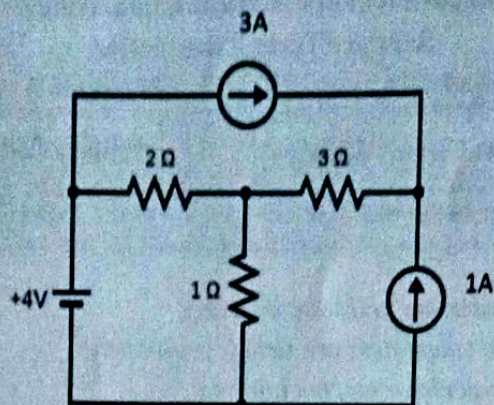
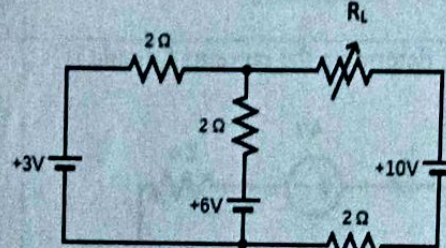
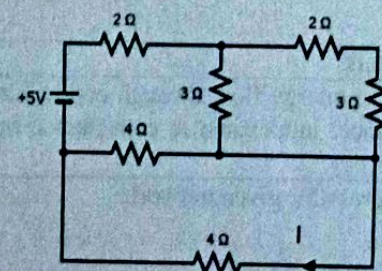
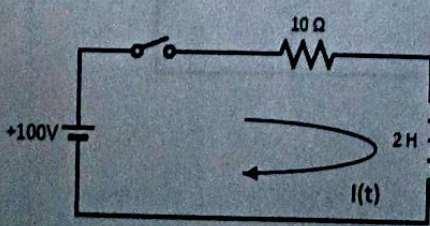
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electrical Engineering
Summer Term Examination
Academic Year: 2024-25 (Semester – I)

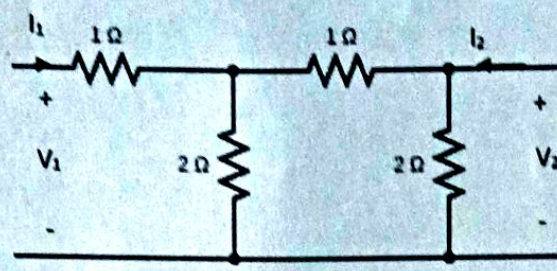
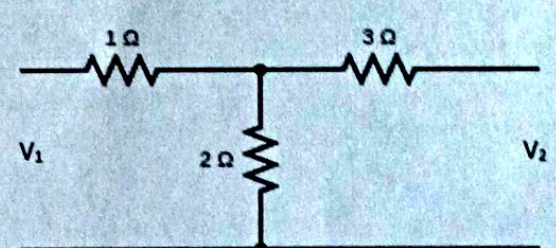
Class: SY B.Tech.	Subject: Circuit Theory	Subject Code: PCC EE-203
Date: 25/06/2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

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- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.
- vi. This paper contains 4 pages

Q.N.	All questions are compulsory		Marks	CO	BT
Q.1	a	Consider the following network, determine the currents i , i_1 and i_2 . 	6	CO1	BT3
	b	Explain the types of sources in detail.	6	CO1	BT2
	c	Explain the mechanism of electrical energy flow through conductor & what is performance of resistance, inductance & capacitance in presence & absence of active sources.	6	CO1	BT2
	d	State and verify the tellegen's theorem for given network. 	6	CO2	BT4

	c	Find current through 1Ω resistor using superposition theorem.	6	CO2	BT4
					
Q.2	a	Find the value of resistance R_L for maximum power transfer and calculate maximum power.	6	CO2	BT4
					
	b	Find the current I and verify reciprocity theorem for the network shown in figure.	6	CO2	BT3
					
	c	In the given circuit switch S is closed at $t=0$. With zero current in the inductor, find i , di/dt and d^2i/dt^2	6	CO3	BT4
					

	d	What is maximum power transfer theorem and prove that source resistance is equal to the load resistance when maximum power is delivered to load.	6	CO2	BT4
	e	What is driving point impedance function, driving point admittance function and voltage transfer function?	6	CO4	BT2
Q.3	a	Find the Z parameters of the network shown in figure. 	6	CO4	BT4
	b	Test whether the following represents driving point immittance function or not. $F(s) = \frac{s^3 + s^2 + 5s + 2}{s^4 + 6s^3 + 9s^2}$ $F(s) = \frac{s^2 + 3s + 2}{s^2 + 6s + 2}$	6	CO4	BT3
	c	Obtain the pole zero plot of following functions. $F(s) = \frac{s(s+2)}{(s+1)(s+3)}$ $F(s) = \frac{s(s+2)}{(s+1+j1)(s+1-j1)}$	6	CO4	BT3
	d	Determine Y parameters for the network shown in figure. 	6	CO4	BT4
	e	What is self-inductance, mutual-inductance and coefficient of coupling?	4	CO4	BT2

Q.4	a	For the network shown in figure determine transfer function V_2/V_1	6	CO4	BT4
	b	For the resistive bridged T network in following figure, find V_2/V_1, V_2/I_1, I_2/V_1 and I_2/I_1.	6	CO4	BT4

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To remember basic concepts and principles of electrical circuits and to solve network problems using mesh current and node voltage equations.
CO 2	To understand and analyse network theorems and their applications.
CO 3	To apply and investigate initial conditions and obtain circuit response using Laplace Transform.
CO 4	To evaluate network functions and two port parameters for electrical network.